Umang Saluja

Ghaziabad, India +91-9568810150 Personal Email College Email
LinkedIn GitHub

Professional Summary

Computer Science undergraduate focused on Machine Learning, Deep Learning, NLP, Computer Vision, and LLM applications. Skilled in Python, Scikit-learn, OpenCV, LangChain, HuggingFace Transformers, RAG pipelines, vector databases, model evaluation, and backend-based ML deployment. Interested in building scalable AI systems and applied ML products.

Education

B.Tech – Computer Science 2024–2028
KIET Group of Institutions
SGPA: 8.61 Till 3rd Semester

Class XII – ISC 2024
94.25%

Class X – ICSE 2022
93.8%

Projects

Flutter Documentation RAG Pipeline Apr 2026 – Present
GitHub Repository

- Built Retrieval-Augmented Generation pipeline using LangChain, ChromaDB, HuggingFace Transformers, and TinyLlama.
- Implemented documentation scraping, data preprocessing, chunking, vector embeddings, and semantic search workflow.
- Used ChromaDB as vector database with scope to extend retrieval using FAISS.
- Ran local LLM inference on Google Colab GPU and experimented with inference optimization.
- Applied prompt engineering to improve contextual response quality and reduce hallucination.

NutriLens – Smart Food Nutrition Analyzer Mar 2026 – Present
GitHub Repository

- Developed AI-powered food classification system using OpenCV, CNN architecture, and Deep Learning workflow.
- Performed image preprocessing, feature extraction, dataset preparation, and model evaluation.
- Trained food classification model using TensorFlow/Keras with scope for PyTorch experimentation.
- Estimated calories, protein, carbohydrates, and fat values after food category prediction.
- Integrated Flask REST APIs for model inference, prediction handling, and mobile app connectivity.

AI Interview Coach App Feb 2026 – Present
GitHub Repository

- Built AI-based interview analysis system using NLP, Scikit-learn, NLTK, and backend ML pipelines.
- Implemented text preprocessing, tokenization, keyword relevance scoring, filler-word detection, and sentiment analysis.
- Designed answer quality, confidence scoring, and performance feedback workflow.
- Integrated FastAPI backend for ML inference, REST API response handling, and Firebase result storage.
- Added scope for HuggingFace Transformer-based answer evaluation and LLM-powered feedback.

Technical Skills

- **Languages:** Python, Java, C
- **Machine Learning:** Scikit-learn, Feature Engineering, Model Evaluation, ML Pipelines
- **Deep Learning:** CNN Architectures, TensorFlow, Keras, PyTorch Basics
- **NLP and LLMs:** LangChain, RAG, Prompt Engineering, HuggingFace Transformers
- **Vector Databases:** ChromaDB, FAISS, Vector Embeddings, Semantic Search
- **Computer Vision:** OpenCV, Image Preprocessing, Food Classification
- **Data Analysis:** Pandas, NumPy, Matplotlib, Seaborn
- **Backend and Deployment:** Flask, FastAPI, REST APIs, MongoDB, Model Deployment, Docker Basics
- **Tools:** Git, GitHub, Jupyter Notebook, Google Colab, VS Code

Coursework

- Machine Learning, Data Analytics, Generative AI
- Natural Language Processing, Computer Vision
- Data Structures and Algorithms, DBMS

Certifications

Machine Learning and Data

- Introduction to Machine Learning Certificate
- Data Analytics with Python Certificate
- NoSQL Databases Certificate
- Oracle SQL Certificate

Backend and Programming

- Flask Certificate
- Django Certificate
- Java OOPS Concepts Certificate

Positions and Achievements

- Member – DSDL, Machine Learning Domain
- Member – Innogeeks, Android Domain
- Participated in Innogeeks Winter of Code 3.0
- First Prize – National Mathematics Day Project Presentation
- Built AI-integrated applications with backend and database connectivity